

Differentiated Carbon Prices or International Transfers?

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the **poorer or greener** a country, the **smaller the abatement burden** it should bear.

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However **efficiency requires a uniform price**, and equity concerns can be settled by **transfers**.

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We compute the emissions rights equivalent to prominent differentiated prices proposals.

Differentiated prices *vs.* a uniform price

Why differentiate? Four imperfections

Absent imperfections, a uniform price with transfers is optimal (Aldy & Stavins, 2012).

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2. Country- or sector-specific distortive taxes (Babiker et al., 2004; Boeters, 2014): if e.g. aviation stays under-taxed, a higher sectoral carbon price improves welfare.

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⇒ **Inconsistent with survey evidence** (Fabre et al., 2025; Fabre, 2026).

None of the four decisively discard a uniform price: 1. uncalibrated, 2. irrelevant, 3. compatible, 4. unproven.

Standard Theory

The static case

Country i produces y_i , consumes c_i , and at carbon price p_i faces abatement cost a_i :

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u_i can be expressed as function of s : we have $u_i(0, p_i) = c_i^a = y_i - a_i$, and $u_i(t, p^*) = c_i^* + t = y_i - a_i^* + t$.

Step 1: the equivalent transfer

$V_i(p_i)$: the lump sum transfer which makes i as well off with the global price as in autarky:

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Optimising country sets $\frac{da_i}{de}(e_i^*) = -p^*$: Therefore, a first-order Taylor expansion around e_i^* gives:

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Equivalent transfer g_i^s of an arbitrary situation $s = (t_i, p_i)$, relative to the benchmark:

$$g_i^s = \underbrace{t_i}_{\text{cash received}} + \underbrace{V_i(p_i)}_{\text{gain from facing } p_i \neq p^*}$$

Step 2: the correspondence between prices and rights

Cap-and-trade at p^* , each person in i endowed with rights r_i . Net monetary transfer to i :

$$t_i^* = (r_i - e_i^*) p^* \quad \implies \quad g_i^* = t_i^*.$$

Autarky, from Step 1: $g_i^a = V_i(p_i) \approx (p^* - p_i) e_i^* \varepsilon^*$. Setting $g_i^* = g_i^a$:

$$\begin{aligned} & \boxed{(r_i - e_i^*) p^* \approx (p^* - p_i) e_i^* \varepsilon^*} \\ \iff & \quad r_i \approx \left(1 + \left(1 - \frac{p_i}{p^*}\right) \varepsilon^*\right) e_i^* \quad \iff \quad p_i \approx \left(1 + \frac{e_i^* - r_i}{e_i^* \varepsilon^*}\right) p^*. \end{aligned}$$

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The key formula for policy design: rights r_i under uniform price equivalent to autarky price p_i

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If the two can always be made equivalent, *is there any reason to prefer one?* Yes.

Step 3: the uniform price weakly dominates, for *every* country

Give country i rights equal to its *autarky* emissions, $r_i = e_i$. Net transfer $\tilde{t}_i = (e_i - e_i^*)p^*$.

Convexity of abatement costs implies:

$$a_i - a_i^* \geq (e_i - e_i^*) \frac{da_i^*}{de} = -(e_i - e_i^*)p^* \quad \implies \quad \tilde{t}_i \geq a_i^* - a_i = V_i(p_i) = g_i^a.$$

Hence, consumption increases *in every country* when trading is allowed and prices equalize:

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This is Diamond & Mirrlees (1971) production efficiency applied to climate: absent other distortions, direct transfers beat differentiated factor prices as a redistributive instrument.

The dynamic case: the equivalence breaks

Add time t ; each variable becomes a trajectory. β_t = discount factor, $R_i = \sum_t r_{it}$ = i 's carbon budget; $G_i = \sum_t g_{it}\beta_t$: total discounted equivalent gain.

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Welfare equivalence now requires equality of *entire discounted streams*, $G_i^U = G_i^A$:

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Special case 1 — homothetic prices, fixed emission rights shares: $p_{it} = \pi_i p_t^*$ and $r_{it} = \rho_i r_t$

$$\mathcal{B} := \sum_t r_t \beta_t p_t^* \quad \rho_i \mathcal{B} - \sum_t e_{it}^* p_t^* \beta_t \approx (1 - \pi_i) \sum_t e_{it}^* \varepsilon_t^* p_t^* \beta_t,$$

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$\Rightarrow$ **No direct correspondence between the intertemporal carbon budget R_i and welfare.**

Special case 1 — homothetic prices, fixed emission rights shares: $p_{it} = \pi_i p_t^*$ and $r_{it} = \rho_i r_t$

$$\mathcal{B} := \sum_t r_t \beta_t p_t^* \quad \rho_i \mathcal{B} - \sum_t e_{it}^* p_t^* \beta_t \approx (1 - \pi_i) \sum_t e_{it}^* \varepsilon_t^* p_t^* \beta_t,$$

Analogous to static formula, but $\mathcal{B}$ still requires the full price path $\Rightarrow$ **not tractable without a model.**

Special case 2: Hotelling

Hotelling rule: Fixed aggregate carbon budget + perfect foresight + efficient capital markets
 $\Rightarrow$ carbon price rises at the rate of return on capital. Discounting at that rate, $\beta_t p_t^* = p_0$ is constant.
 $\beta_t p_t^*$ then factors out of both sums, and the dynamic condition collapses to:

$$R_i - E_i^* \approx (1 - \pi_i) \sum_t e_{it}^* \varepsilon_t^* \quad E_i^* = \sum_t e_{it}^*$$

Structurally identical to the static formula: the intertemporal carbon budget R_i plays the role of the single-period allocation, and the homothetic price factor π_i the role of p_i/p^* .

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In the numerical analysis, we place ourselves in the Hotelling case.

Numerical analysis

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The formula rests on approximations a calibrated model need not satisfy: homogeneous elasticities, homothetic price paths, a first-order Taylor expansion.

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Grid search over (ρ_i, π_i) for eight major regions: **USA, Russia, EU27, China, Turkey, India, Nigeria, DRC**.

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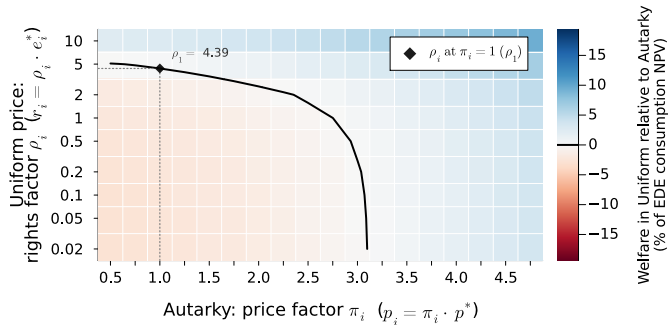
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RoW receives rights so that global emissions coincide with autarky, grandfathered on baseline emissions.

Reading the heatmaps

USA welfare **gain of uniform over autarky**, depending on rights and price.

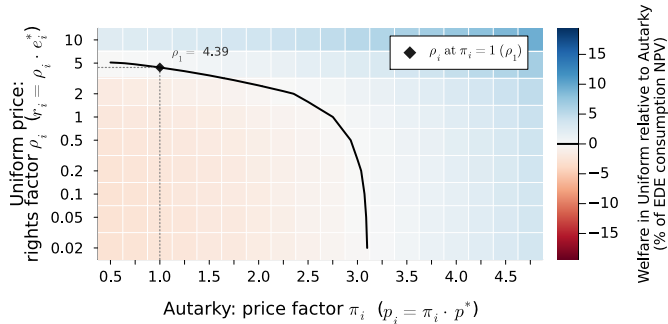


Grid showing the welfare **gain** or **loss** of *Uniform over Autarky*

Blue: Uniform price with transfers dominates; **Red:** Autarky dominates; line: indifference curve.

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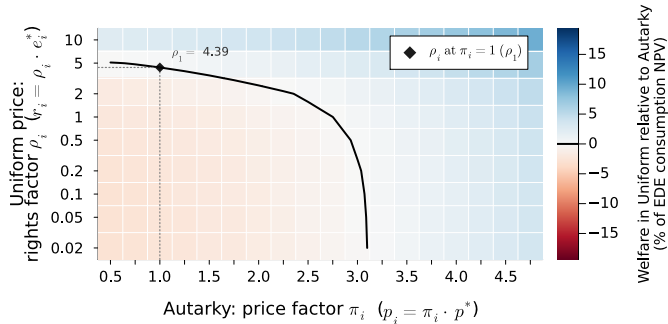
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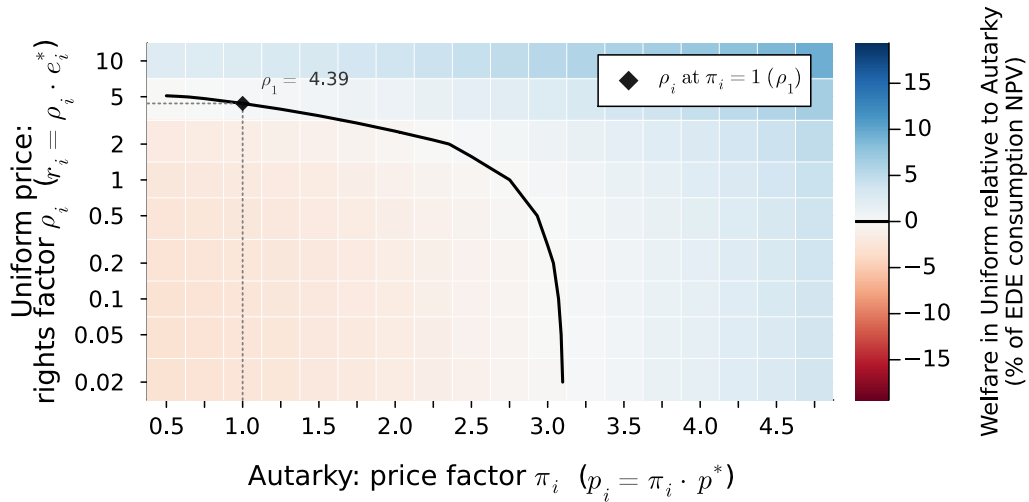
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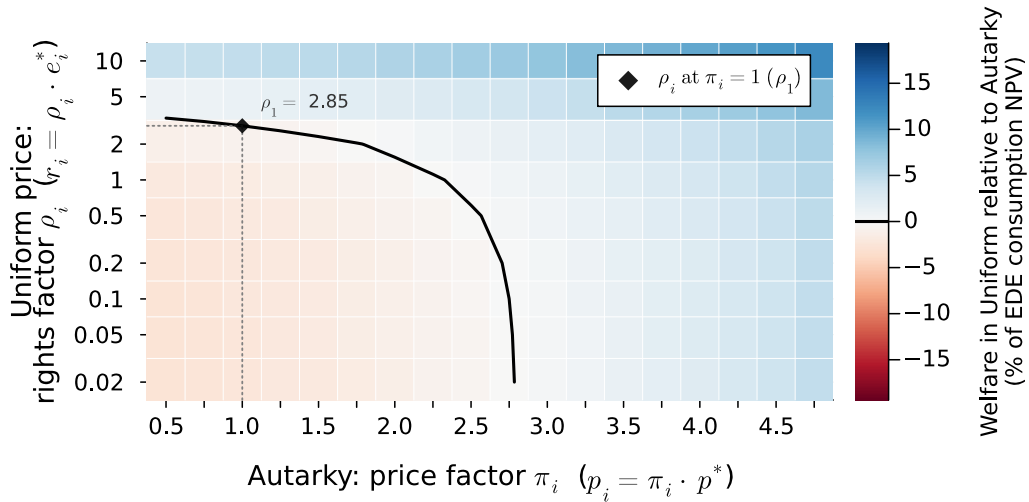
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Reading: The U.S. would need at least $\rho_1 = 4.39$ times more rights p.c. than the world average to be as well as off as in autarky, with the same global price.

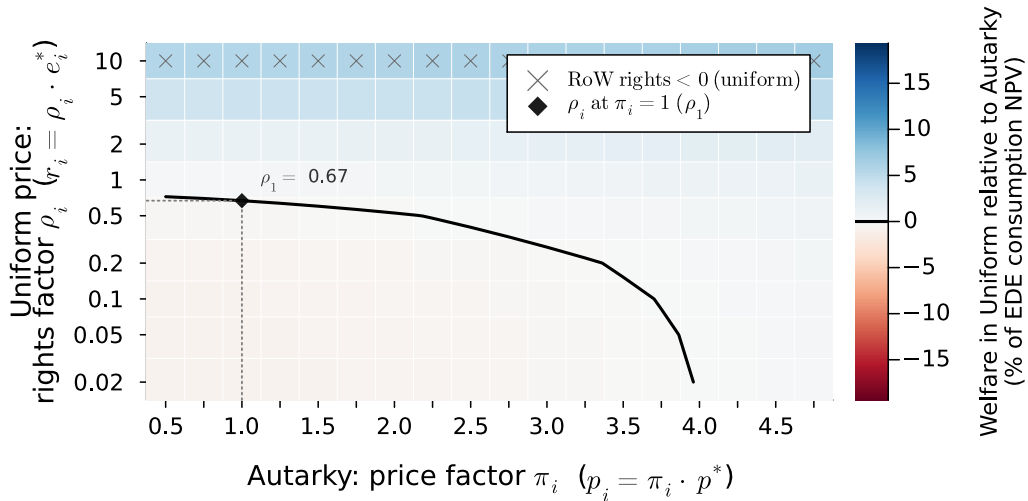
United States



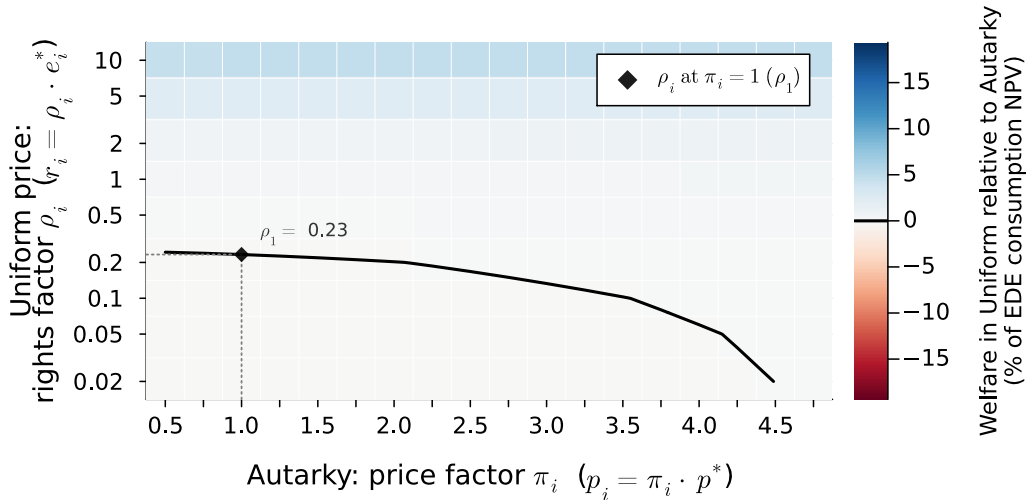
Russia



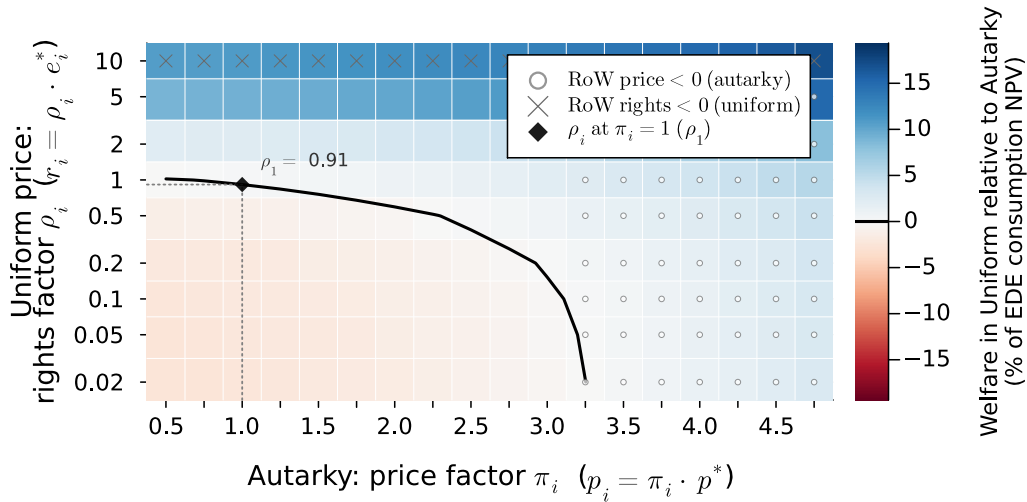
China



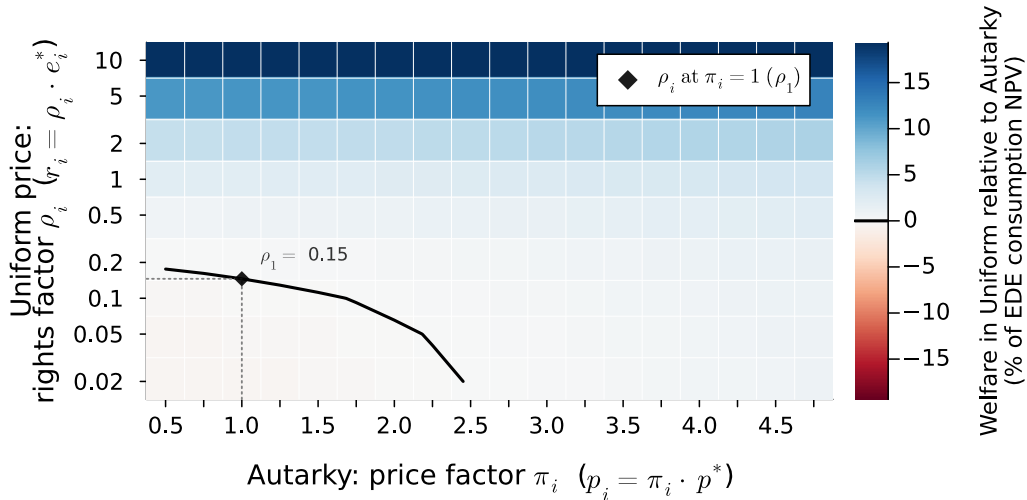
European Union (EU27)



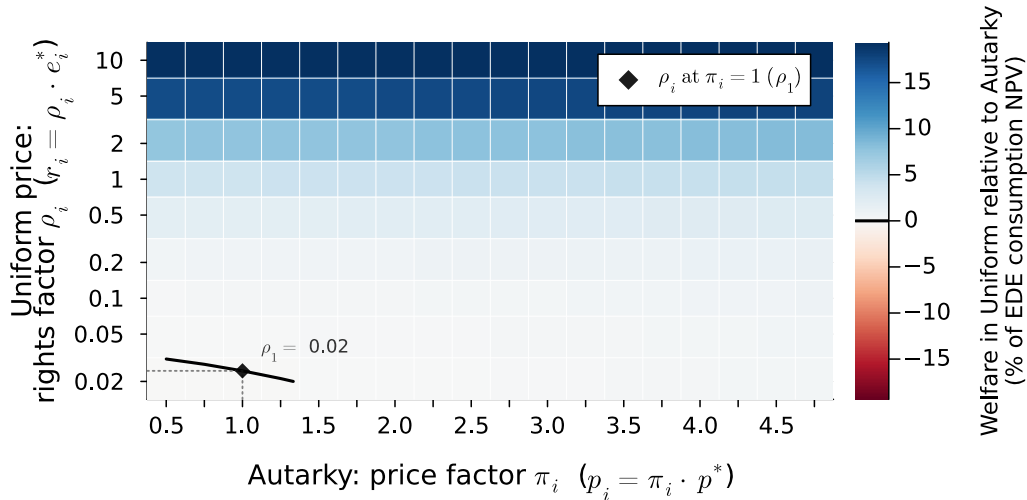
India



Nigeria



Democratic Republic of the Congo



Assessment of recent proposals

The proposals

IMF; Working Group on Climate Coalitions propose a coalition with price floors differentiated by income: \$75/tCO₂ high-income, \$50/t upper-middle, \$25/t lower-income (Parry et al., 2021; Wolfram et al., 2025).

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Banerjee, Duflo & Greenstone (2025) propose a grand bargain with \$10 LICs, \$30 lower-MICs, \$50 upper-MICs (no price asked of HICs, I assume \$75/t).

Pricing gives right to climate-damage compensation, funded by HICs through global taxes on wealth and profit.

Equivalent rights for the proposals

Country	Wolfram et al.			Banerjee et al.			Equal Right, 5%/yr		
	p_i	ρ_i^{EDE}	ρ_i^{cons}	p_i	ρ_i^{EDE}	ρ_i^{cons}	p_i	ρ_i^{EDE}	ρ_i^{cons}
United States	0	—	—	75	2.82	3.06	240	0.83	1.22
Russia	0	—	—	75	1.89	1.98	240	0.6	0.77
European Union	75	0.36	0.38	75	0.36	0.36	232	0.38	0.4
Turkey	0	—	—	50	1.62	1.57	108	1.45	1.38
Indonesia	50	0.92	1.0	50	1.04	1.06	96	0.91	0.91
China	50	0.98	0.96	50	0.95	0.91	108	1.31	1.23
Brazil	50	0.37	0.46	50	0.41	0.47	108	0.36	0.42
India	25	1.12	1.16	30	1.03	1.07	42	1.06	1.07
Nigeria	0	—	—	30	0.11	0.17	42	0.13	0.18
DR Congo	0	—	—	10	0.01	0.03	15	0.01	0.03

Variant 1: the equivalent rights as solved, the allocation sets the cap

World temp. 2100, change (°C)	—0.019	—0.013	—0.034	—0.026	—0.043	—0.040
Emissions change in the coalition (%)	—9.5%	—6.6%	—8.5%	—6.4%	—17.4%	—16.3%
World welfare gain, EDE (%)	0.052%	0.039%	0.068%	0.053%	0.114%	0.110%
World consumption gain (%)	0.030%	0.028%	0.036%	0.035%	0.077%	0.086%

Variant 2: rights raised to the coalition's emissions, shared to maximise the smallest gain

World welfare gain, EDE (%)	0.018%	0.009%	0.096%	0.045%	0.211%	0.138%
World consumption gain (%)	0.021%	0.020%	0.056%	0.052%	0.187%	0.182%

p_i : the price the proposal asks of i in 2030 (\$/t), against one club price for all coalition members of \$49.2/t (Wolfram et al.), \$55.8/t (Banerjee et al.) and \$138.5/t (Equal Right, 5%/yr). ρ_i^{EDE} and ρ_i^{cons} hold each member at, respectively, its equally-distributed-equivalent and its mean consumption per capita.

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Still one key reason for differentiation: **gradual transition**.

Thank you for your attention!

bit.ly/uniform_price

adrien.fabre@cnr.fr

Appendix

Two boundary conditions evaluated in *every* year of 2030–2100:

1. **Market-distorting subsidies** ($\omega_i \times \pi_i > 1$): the target country claims more than 100% of the global price burden. To compensate, the model is forced to push $p_{-i} < 0$ — economically, a **fossil-fuel subsidy for the RoW**.

2. **Negative emission rights** ($\rho_i > 1 / pop_i$): granting i its multiplier uses more than the whole global budget, forcing the RoW into **negative rights**. Binds for large populations: countries above 10% of world population (China, India) trigger it for any $\rho_i > 10$.

RoW price identity, the starting point of the root-finding:

$$p^* = \omega_i p_i + (1 - \omega_i) p_{-i}, \quad \omega_i = e_i / E^* \quad \implies \quad p_{-i} = p^* (E^* - e_i \pi_i) / e_{-i}.$$

This linear approximation assumes emission-weighted price averaging delivers the same global abatement — false with non-linear MACs, hence the **Newton–Raphson** correction to $< 0.1\%$ relative error on the cap.

Autarky scenario, United States ($\pi = 2$) [Go back](#)

Variable	2030	2040	2050	2060	2070	2080	2090	2100
p^* (USD/tCO ₂)	42.6	112.9	205.9	285.7	401.6	480.9	476.2	471.4
p_i (USD/tCO ₂)	85.2	225.8	411.7	571.4	803.3	961.9	952.3	942.8
E_i (GtCO ₂)	4.31	2.58	0.63	-0.53	-1.75	-2.22	-1.86	-1.52

Net-negative US emissions from the 2060s: abatement rates $\mu_i > 1$, i.e. [CDR deployment](#), with the MAC function extrapolated beyond the 100% boundary.

Standard IAMs constrain $\mu_i \in [0, 1]$; enforcing that bound crashes the grid search under extreme policy parameters.

The equivalent price, and why we do not use it [▶ Go back](#)

One could instead define an **equivalent price** p_i^* such that $(0, p_i^*)$ is welfare-equivalent to (t_i, p_i) :

$$t_i + (p^* - p_i)e_i^*\varepsilon^* \approx (p^* - p_i^*)e_i^*\varepsilon^* \quad \implies \quad p_i^* \approx p_i - \frac{t_i}{e_i^*\varepsilon^*}.$$

Problem: p_i^* can be *negative* when the transfer is large, which has no natural interpretation as a carbon price.

The equivalent transfer g_i^s does not share this limitation $\implies$ we use it as the primary welfare metric.

Last slide (needed)